

Software Engineering

CS-2105

Government College University Lahore

Assignment: Developing the Software Requirements Specification

SRS

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1 Introduction

1.1 Purpose

This document outlines the software requirements for a system designed to enhance the efficiency and reliability of power transmission at the National Transmission & Dispatch Company (NTDC). The system will address challenges posed by outdated infrastructure and inadequate maintenance, aiming to reduce transmission losses and improve the stability of electricity supply across Pakistan.

1.2 Scope

In Scope	Out of Scope
Real-time monitoring of transmission lines and equipment status	Advanced machine learning-based predictive analytics
Predictive maintenance alerts to prevent equipment failures	Highly customizable reports
Data analytics to identify high-loss areas and inefficiencies	Integration with external data sources
Automated reporting for stakeholders	
Integration with existing NTDC infrastructure for data collection and system integration	

1.2.1 In Scope

The software will focus on enhancing the efficiency and reliability of power transmission by incorporating real-time monitoring, predictive maintenance, and data analytics. It will integrate with existing NTDC infrastructure for data collection and system integration.

1.2.2 Out Scope

Advanced machine learning-based predictive analytics, highly customizable reports, and integration with external data sources are outside the scope of this phase.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
NTDC	National Transmission & Dispatch Company
DISCOs	Distribution Companies
SRS	Software Requirements Specification
RTM	Requirements Traceability Matrix
MOSCOW	Must have, Should have, Could have, Won't have
UI	User Interface

1.4 References

1. IEEE 830-1988 (or its modern equivalent, IEEE 29148)
2. NTDC internal documentation
3. Meeting with Federal Minister for Power, Laghari, September 21, 2024

1.5 Overview

This document details the requirements for a software system designed to improve the efficiency and reliability of power transmission at NTDC. It covers the overall description of the system, specific functional and non-functional requirements, use cases, stakeholder feedback, visual representations of the system, and a requirements traceability matrix.

2 Overall Description

2.1 Product Perspective

- **Key Role in Modernization:** The software system will be integral to modernising NTDC's power transmission infrastructure.
- **System Integration:** It seamlessly integrates with existing systems to enhance functionality.
- **Real-Time Monitoring:** Provides real-time monitoring of the power transmission network.
- **Predictive Maintenance:** Incorporates predictive maintenance capabilities to address potential issues proactively.
- **Data Analytics:** Offers advanced data analytics for informed decision-making.
- **Reduced Transmission Losses:** Helps in minimising transmission losses through efficient operations.

- **Improved Stability:** Enhances the overall stability and reliability of the power grid.

2.2 Product Features

1. Real-time monitoring of transmission lines and equipment status.
2. Predictive maintenance alerts to prevent equipment failures.
3. Data analytics to identify high-loss areas and inefficiencies.
4. Automated reporting for stakeholders.

2.3 User Classes and Characteristics

2.3.1 Primary Users

- **Operations staff** responsible for managing daily tasks in power transmission.
- **System operators** who monitor the network in real-time.
- **Maintenance teams** who use predictive tools to identify potential problems.

2.3.2 Secondary Users

- **Government regulators** who use reports to ensure NTDC meets service standards.
- **Stakeholders** who rely on data for decision-making.
- **Consumers** who benefit from a more reliable power supply.

2.4 Operating Environment

The software will operate within the existing NTDC IT infrastructure. This includes:

- **Hardware:** Servers, workstations, and communication devices.
- **Software:** Operating systems, database management systems, and existing monitoring tools.
- **Network:** A secure and reliable network connecting all components.

2.5 Assumptions and Dependencies

- **NTDC** will provide the necessary data and access to existing systems for integration.
- **Adequate training** will be provided to NTDC staff on how to use the new software.
- **Regular maintenance and updates** will be performed to ensure the system's continued effectiveness

3 Specific Requirements

3.1 Functional Requirements

3.1.1 Real-time Monitoring

- The system shall continuously **track the performance** of transmission lines and equipment.
- The system shall provide **real-time updates** on the status of key components.
- The system shall generate **alerts** for any anomalies or critical events detected.

3.1.2 Predictive Maintenance

- The system shall analyze **historical and real-time data** to predict potential equipment failures.
- The system shall generate alerts for **upcoming maintenance** needs.
- The system shall allow users to **schedule and track** maintenance activities.

3.1.3 Data Analytics

- The system shall **analyze transmission data** to identify areas with high energy loss.
- The system shall provide **reports and visualizations** to highlight inefficiencies.
- The system shall allow **users to perform custom queries and analysis** on the data.

3.1.4 Automated Reporting

- The system shall generate regular reports on transmission performance, outages, and maintenance needs
- The system shall allow users to customize report formats and content.

3.2 Non-functional Requirements

3.2.1 Performance

- The system shall respond to user requests within 5 seconds.
- The system shall be able to handle a peak load of 10,000 concurrent users.
- The system shall have an uptime of 99.9%.

3.2.2 Usability

- The system shall have an intuitive user interface that is easy to navigate.
- The system shall provide clear and concise instructions for all functions.
- The system shall be accessible to users with disabilities.

3.2.3 Security

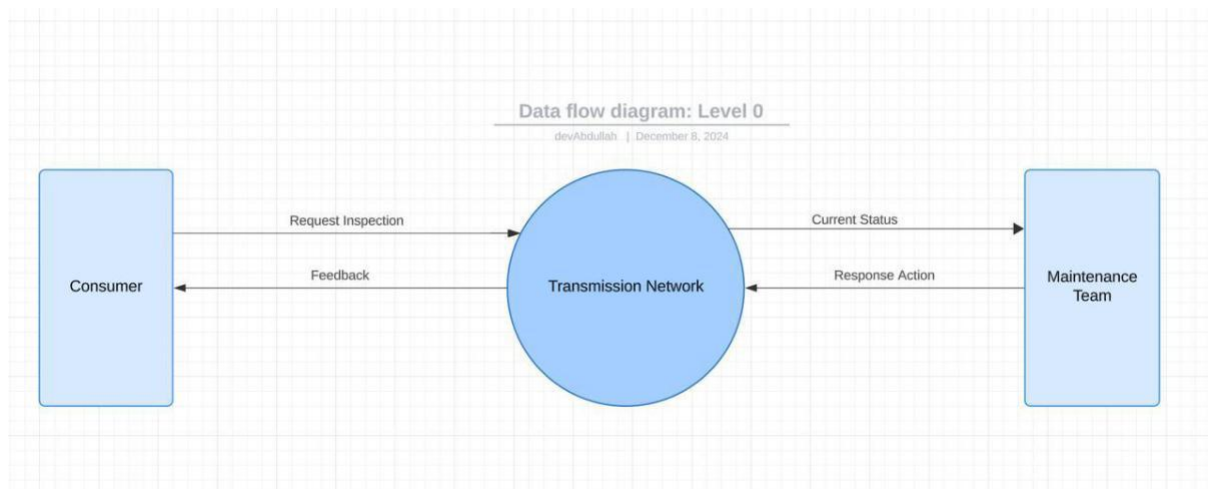
- The system shall protect sensitive data from unauthorized access.
- The system shall comply with all relevant security standards and regulations.
- The system shall have a robust backup and recovery mechanism.

3.2.4 User-Friendliness

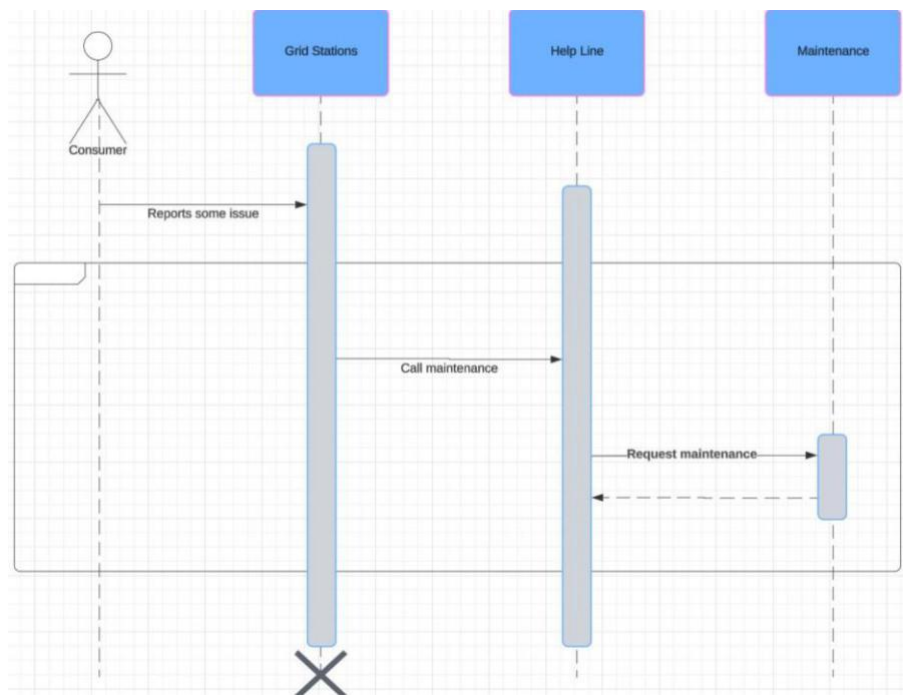
- Intuitive UI for NTDC operators.

3.3 Defining Requirements Using Diagrams

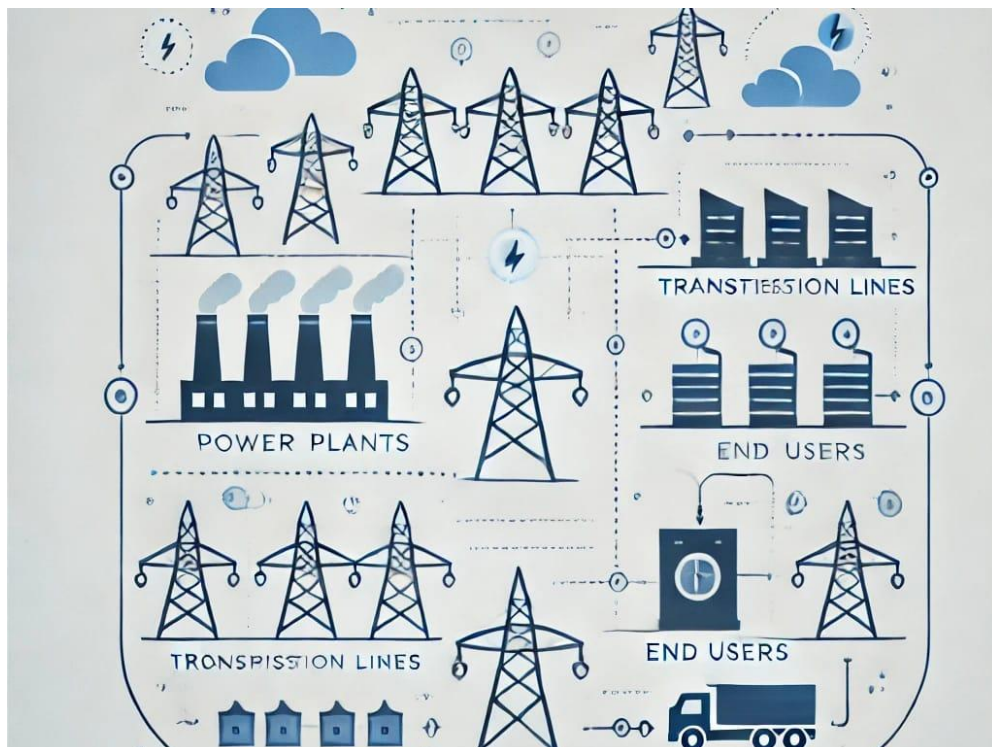
3.3.1 Data flow diagram



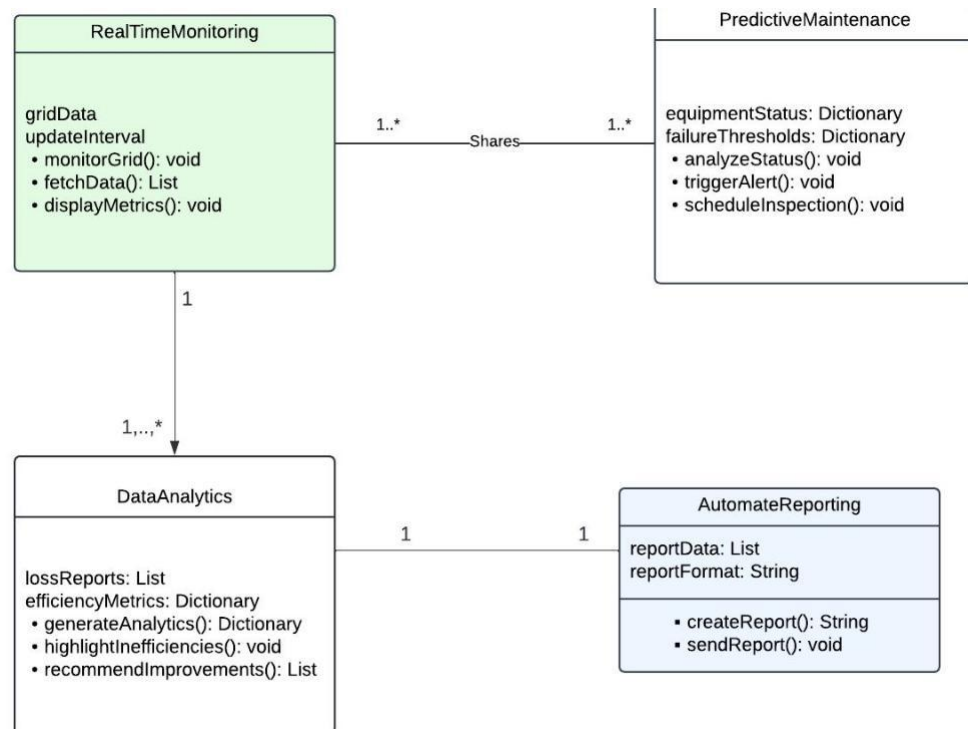
3.3.2 Sequence Diagram



3.3.3 Contribution Diagram



3.3.4 Class Diagram



3.4 External Interface Requirements

3.4.1 User Interfaces

- The system shall provide a web-based user interface for accessing and managing the system.
- The system shall provide mobile access for authorized users.

3.4.2 Hardware Interfaces

- The system shall integrate with existing sensors and monitoring devices.
- The system shall support data acquisition from various communication protocols.

3.4.3 Software Interfaces

- The system shall integrate with existing NTDC databases and applications.
- The system shall provide APIs for data exchange with other systems.

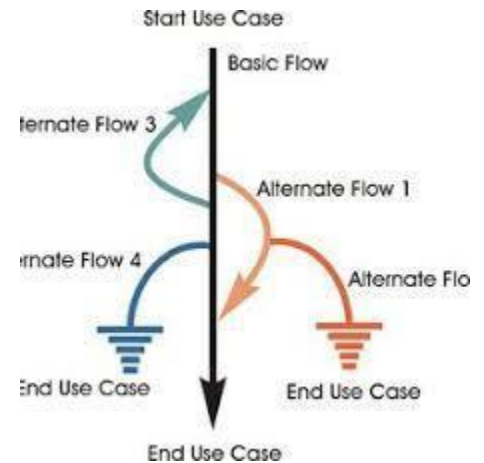
3.4.4 Communication Interfaces

- The system shall support secure communication protocols for data transmission.
- The system shall provide notifications and alerts through email and SMS

4 Use Cases

4.1 Use Case 1: Proactive Equipment Maintenance

- Brief Description: The system predicts potential equipment failure and notifies the maintenance team to perform proactive repairs.
- Actors: System Operator, Maintenance Team
- Basic Flow:
 1. The system analyzes real-time and historical data to predict potential equipment failure.
 2. The system generates an alert notifying the system operator of the potential failure.
 3. The system operator reviews the alert and acknowledges receipt.
 4. The system automatically notifies the maintenance team.
 5. The maintenance team schedules and performs the necessary repairs.
- Alternative Flows:
 - If the system operator does not acknowledge the alert within a certain time frame, the system escalates the notification.
 - If the maintenance team cannot perform the repairs immediately, they can reschedule the maintenance activity within the system.



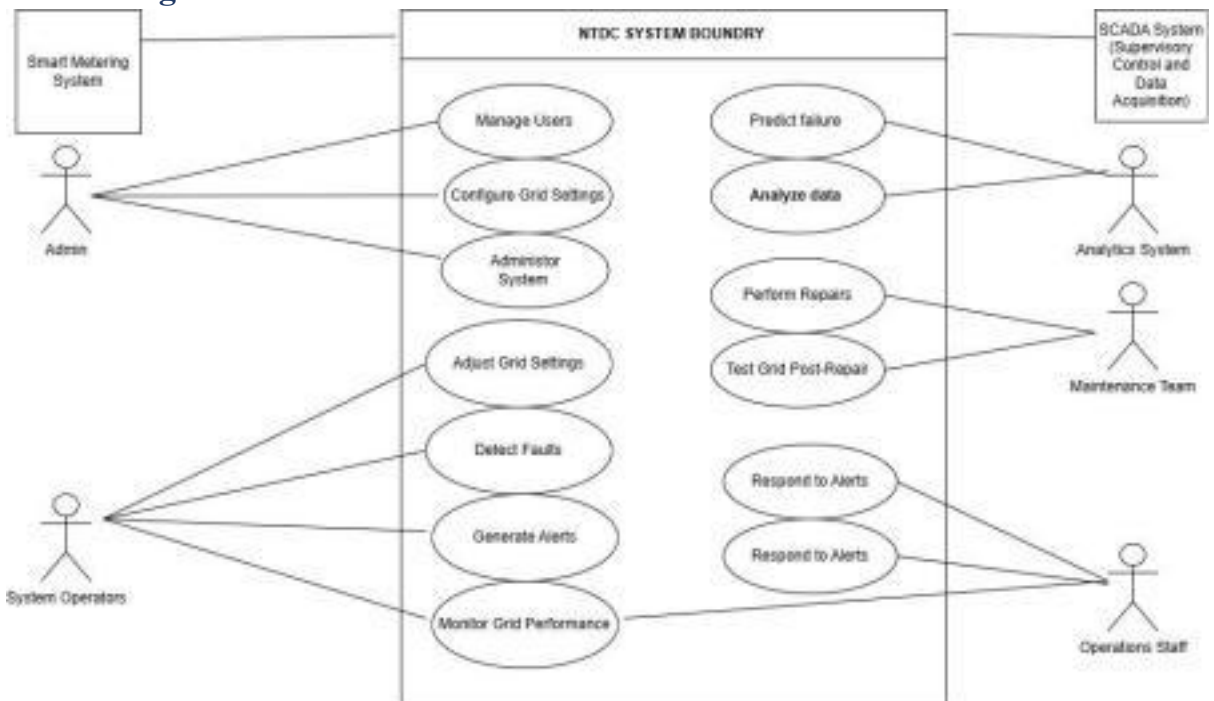
4.2 Use Case 2 : Respond to Grid Event

- **Brief Description:** The system detects a fault or anomaly in the power grid and alerts the appropriate personnel to take corrective action.
- **Actors:** System Operators, Operations Staff, Maintenance Team (depending on the nature of the event)
- **Basic Flow:**
 1. Sensors and monitoring devices throughout the grid continuously collect data and transmit it to the system.
 2. The system analyzes the data in real-time, looking for patterns and anomalies that indicate a potential problem (e.g., voltage fluctuations, line overloads, equipment malfunctions).
 3. If a fault or anomaly is detected, the system generates an alert with details about the event, including its location, severity, and potential impact.
 4. The system notifies the appropriate personnel based on the nature of the event:
 - System Operators are notified of all events and can view detailed information on dashboards and reports.
 - Operations Staff are notified of events that require immediate attention or field intervention.
 - Maintenance Teams are notified of events that indicate potential equipment failures or require scheduled maintenance.
 5. The notified personnel acknowledge the alert and take appropriate action to address the event, such as:
 - System Operators may adjust grid settings or reroute power flow to stabilize the network.
 - Operations Staff may be dispatched to the site to investigate and resolve the issue.
 - Maintenance Teams may schedule and perform repairs or preventative maintenance on the affected equipment.

- **Alternative Flows:**

- If an alert is not acknowledged within a specified timeframe, the system escalates the notification to higher-level personnel.
- The system allows users to customize alert thresholds and notification preferences based on their roles and responsibilities.
- The system can automatically trigger pre-defined procedures or actions in response to certain types of events.

4.3 Use case diagram



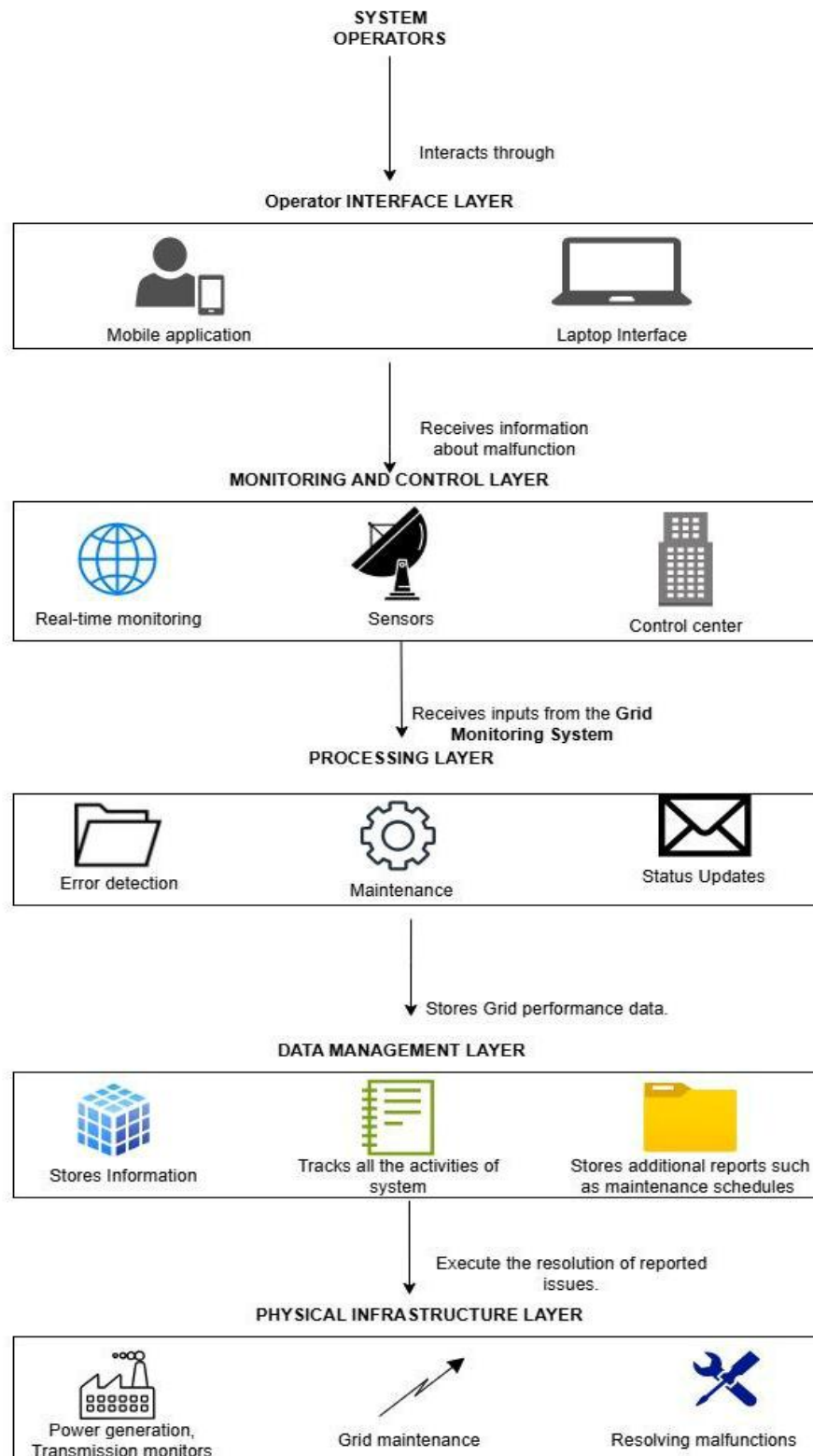
5 Feedback

Feedback was collected from NTDC staff, government regulators, and industry experts to refine the requirements and ensure the system meets all stakeholders' needs. Key feedback and its resolution include:

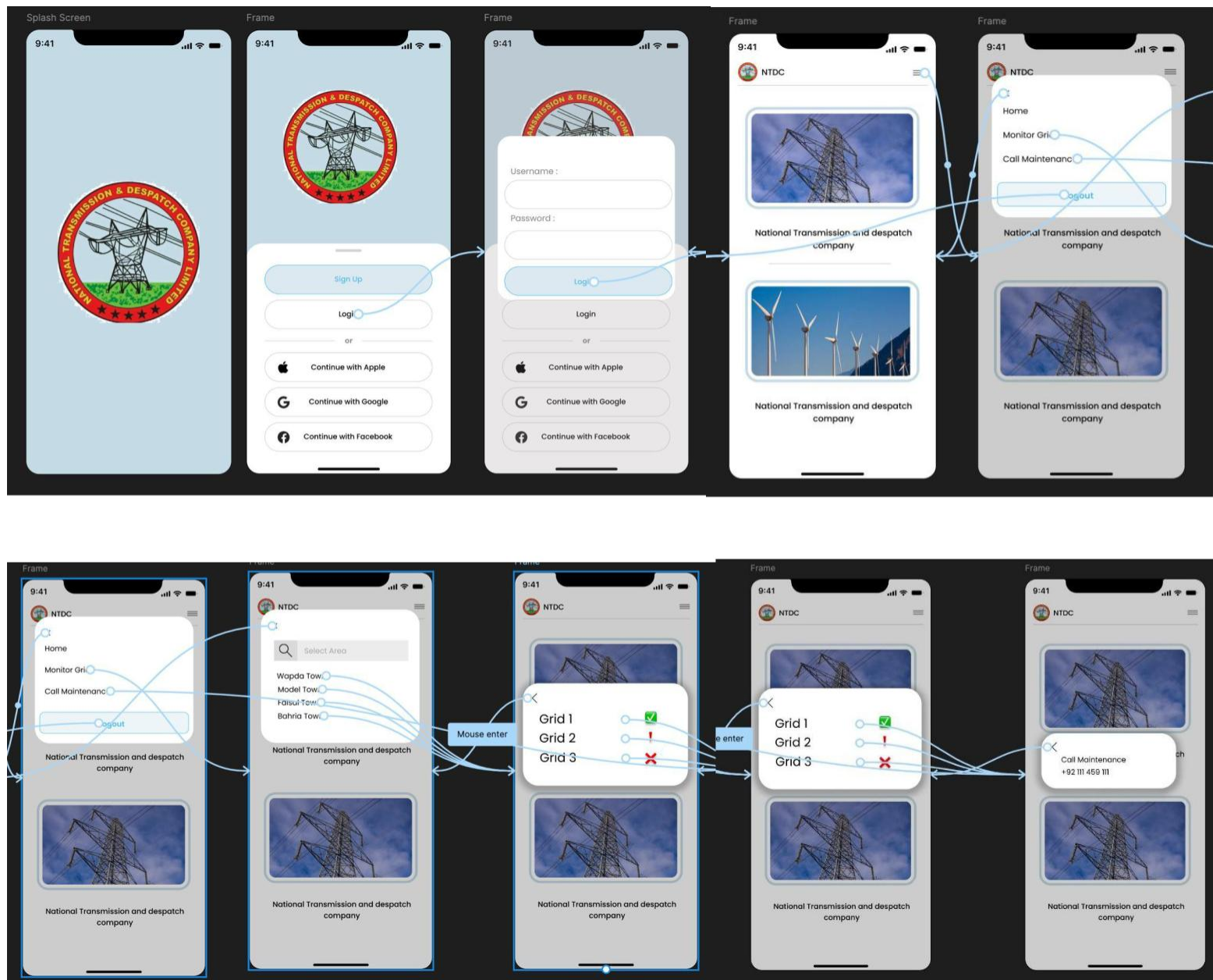
- **NTDC staff** emphasized the need for an intuitive interface and comprehensive training materials, leading to usability testing and the development of detailed user documentation.
- **Government regulators** emphasized data security and compliance, prompting the implementation of strong security measures and adherence to industry standards.
- **Industry experts** recommended incorporating advanced analytics, which was addressed by adding features for data analysis, visualization, and custom reporting.

6 Analysis Models

6.1 System Architecture Diagram



6.2 User Interface Wireframes



7 Determined List

R ID	Requirement Description	A1: Tools	A2: Problem	A3: Vision	A4: Scope	Final: Interview
R01	Real-time monitoring of transmission lines and equipment	✓	✓	✓	✓	✓
R02	Predictive maintenance alerts to prevent equipment failures	✓	✓	✓	✓	✓
R03	Data analytics to identify high-loss areas and inefficiencies	✓	✓	✓	✓	✓
R04	Automated reporting for stakeholders			✓	✓	
R05	Integration with existing NTDC infrastructure		✓	✓	✓	✓
R06	User-friendly interface for NTDC operators	✓		✓		✓
R07	Data security and access control		✓	✓		✓
R08	Scalability to handle future grid expansion			✓	✓	
R09	High reliability and uptime			✓	✓	✓
R10	Addressing outdated infrastructure		✓	✓	✓	✓
R11	Managing overloaded transmission lines		✓		✓	✓
R14	Training and skill development for staff	✓	✓		✓	✓